Exp No:5

Data Collection and Initial Exploration

AIM:

To collect, load, and perform initial exploration of the diabetes dataset.

Materials Needed:

Diabetes dataset (available from sources such as UCI Machine Learning Repository or Kaggle)
Python with libraries: pandas, numpy, matplotlib, seaborn

PROGRAM:

```
import pandas as pd
import csv
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('diabetes.csv')
print(df.head())

print(df.info())
print(df.describe())

df.hist(bins=50, figsize=(20,15))
plt.show()

sns.pairplot(df)
plt.show()
```

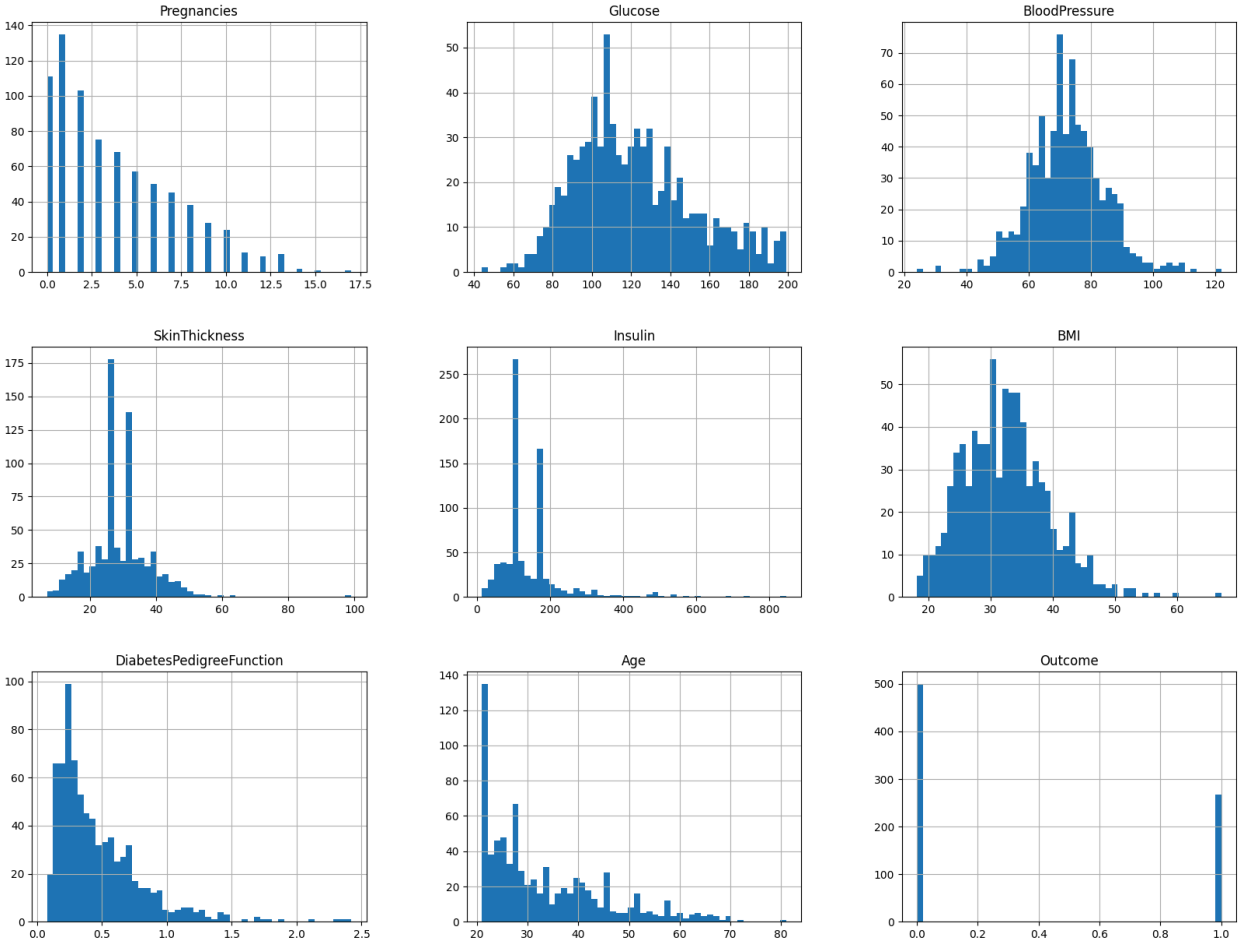
OUTPUT:

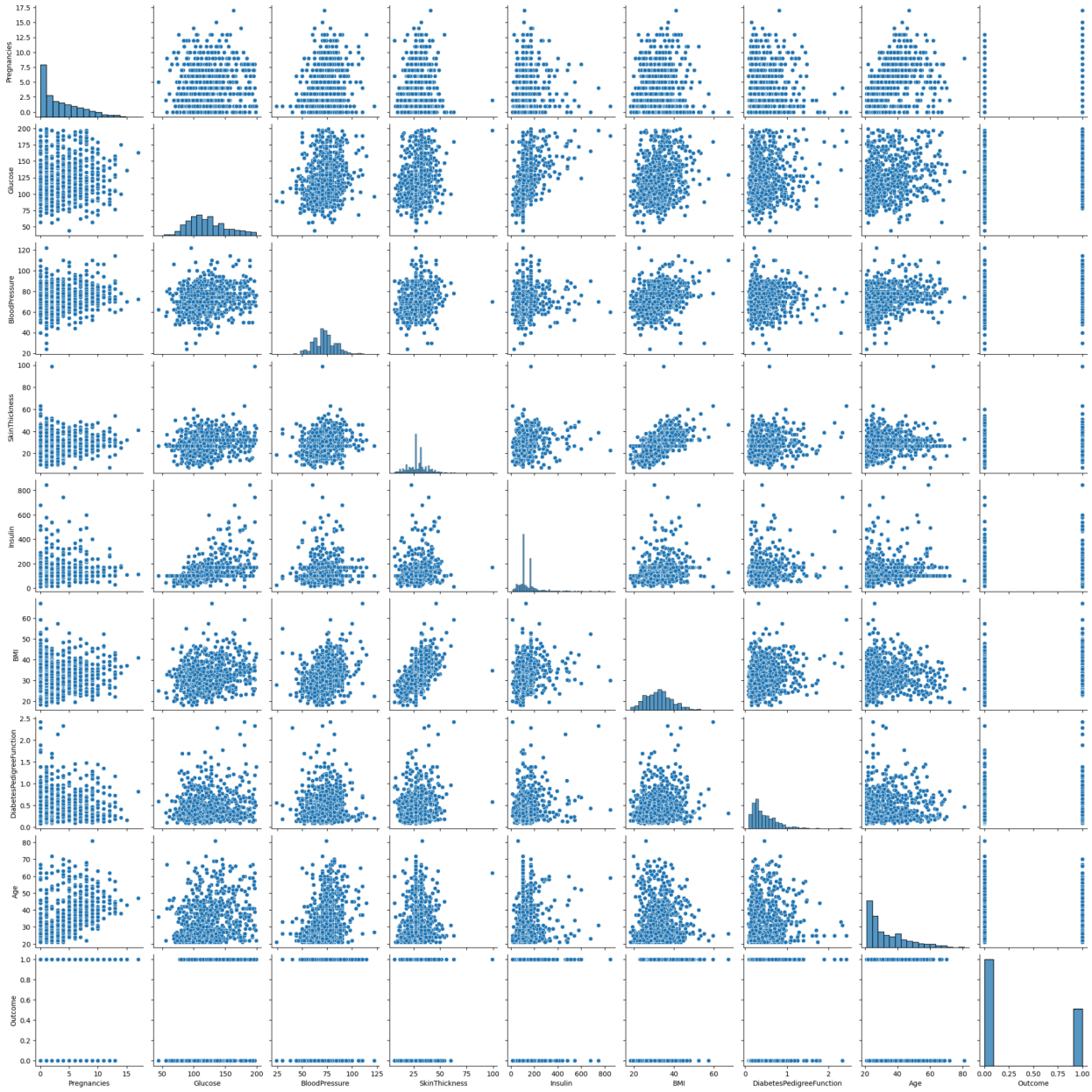
```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148           72.0             35    169.5  33.6
1            1       85           66.0             29    102.5  26.6
2            8      183           64.0             32    169.5  23.3
3            1       89           66.0             23     94.0  28.1
4            0      137           40.0             35    168.0  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
```

```
2                  0.672   32       1
3                  0.167   21       0
4                  2.288   33       1
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    float64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    float64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(4), int64(5)
memory usage: 54.1 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      3.845052  121.677083      72.389323      29.089844  141.753906
std       3.369578   30.464161      12.106039       8.890820   89.100847
min       0.000000   44.000000      24.000000       7.000000   14.000000
25%       1.000000   99.750000      64.000000      25.000000  102.500000
50%       3.000000  117.000000      72.000000      28.000000  102.500000
75%       6.000000  140.250000      80.000000      32.000000  169.500000
max      17.000000  199.000000     122.000000      99.000000  846.000000

              BMI  DiabetesPedigreeFunction         Age     Outcome
count  768.000000               768.000000  768.000000  768.000000
mean    32.434635                 0.471876   33.240885    0.348958
std      6.880498                 0.331329   11.760232    0.476951
min     18.200000                 0.078000   21.000000    0.000000
25%     27.500000                 0.243750   24.000000    0.000000
50%     32.050000                 0.372500   29.000000    0.000000
75%     36.600000                 0.626250   41.000000    1.000000
max     67.100000                 2.420000   81.000000    1.000000
```


Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome



RESULT:

The experiment has been carried out successfully.